

EQUINESYNC PRODUCT IMPLEMENTATION ATLAS

MASTER STANDARD AND CONTROLLED TEMPLATE

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Related Implementation Authority: EquineSync MAIP

Repository of Record: <https://github.com/rianray2012-coder/EquineSync-V4.git>

PART I

MASTER STANDARD

1. Purpose

This Master Standard establishes the mandatory operating model, content structure, quality requirements, lifecycle controls, drafting rules, review procedures, implementation gates, traceability requirements, evidence standards, and reusable template for every EquineSync Product Implementation Atlas.

A Product Implementation Atlas, or PIA, is the controlled domain-level bridge between EquineSync's governing decisions and executable software implementation.

The controlling implementation chain is:

Constitutional Governance → Founder Decisions → MAIP → Product Implementation Atlas → Engineering Work Packages → Code and Configuration → Tests → Evidence → Release Decision → Controlled Enrollment

This Standard prevents EquineSync from being implemented through:

- undocumented assumptions;
- disconnected feature lists;
- informal conversations;
- inconsistent AI-generated specifications;
- engineering convenience decisions;
- competing sources of truth;
- unapproved scope expansion;
- incomplete security or permission models;
- demonstrations mistaken for verification; or
- release decisions unsupported by evidence.

Every PIA must enable EquineSync to determine:

1. What must be built.
2. Why it must be built.
3. What is explicitly not being built.
4. Who may use each capability.
5. Which relationships authorize each action.
6. Which data the capability owns, references, derives, or displays.
7. How each workflow behaves from initiation through completion.
8. What happens when the workflow fails, conflicts, or is interrupted.
9. How the capability will be implemented.
10. How the implementation will be tested.
11. What evidence will prove conformance.
12. How the capability will be deployed, supported, monitored, recovered, and maintained.
13. Whether the capability is safe and complete enough for first-user enrollment.

2. Controlling Definition

The Founder has adopted the following definition:

A Product Implementation Atlas is the authoritative, version-controlled, domain-specific implementation instrument that translates EquineSync constitutional governance, MAIP requirements, founder decisions, product workflows, operational rules, data responsibilities, technical contracts, verification requirements, and release conditions into a complete and traceable basis for software construction and assurance.

This definition controls the drafting, review, approval, implementation use, validation, maintenance, revision, and supersession of all EquineSync PIAs.

3. Aligned Purpose

Every PIA must ensure that implementation is:

- constitutionally compliant;
- operationally realistic;
- technically coherent;
- securely designed;
- privacy-aware;
- safeguarding-aware;
- objectively testable;
- evidence-supported;
- supportable;
- recoverable;
- maintainable;
- properly scoped; and
- capable of supporting a controlled enrollment decision.

A PIA is not merely a product requirements document, technical design, interface mockup, backlog, database model, or test plan.

It is the controlled integration of all those elements for an identified capability or domain.

4. Normative Language

Term	Meaning
MUST / SHALL	Mandatory requirement. Failure blocks the applicable gate unless a controlled deviation is approved.
MUST NOT / SHALL NOT	Prohibited action or condition.
SHOULD	Strong recommendation. Departure requires recorded reasoning.
SHOULD NOT	Strong presumption against the action.
MAY	Permitted but not required.
CONTROLLED	Subject to defined authority, versioning, review, evidence, and change control.
AUTHORITATIVE	The recognized source of truth for the identified matter.
MATERIAL	Capable of affecting functionality, authority, safety, security, privacy, data integrity, legal posture, implementation scope, cost, supportability, or release readiness.
DESIGN-COMPLETE	Product and implementation requirements are sufficiently defined for structured review.

Term	Meaning
IMPLEMENTATION-READY	Engineering can proceed without making unauthorized material decisions.
IMPLEMENTED	The approved capability has been constructed in the identified environment and code baseline.
CONFORMANT	The as-built implementation matches the approved PIA or an authorized change.
VERIFIED	Objective tests and preserved evidence demonstrate satisfaction of the applicable requirements.
OPERATIONALLY READY	Deployment, monitoring, support, recovery, correction, and maintenance controls are ready.
FIRST-USER ENROLLMENT	Controlled admission and onboarding of the first non-founder user, organization, facility, trainer, owner, rider, guardian, staff member, or other external participant into an operational EquineSync environment.

Ambiguous terms such as “easy,” “secure,” “fast,” “intuitive,” “appropriate,” “as needed,” or “normally” must not be used as standalone requirements.

When such terms are necessary, they must be paired with measurable criteria or a defined decision rule.

5. Authority and Order of Precedence

The order of controlling authority is:

1. Locked EquineSync constitutional governance
2. Founder-adopted or founder-approved decisions
3. Applicable constitutional companion artifacts and authoritative registers
4. This PIA Master Standard
5. The controlling MAIP
6. Approved foundational PIAs
7. Approved domain and cross-domain PIAs
8. Approved implementation decisions and deviations
9. Engineering work packages
10. Source code, infrastructure, schemas, migrations, and configuration
11. Test plans and validation scripts
12. Operational procedures, support materials, and training content

A lower-authority artifact must not override a higher-authority artifact.

When a conflict is identified:

- the conflict must be recorded;

- the affected requirement must be classified;
- work must pause when continued implementation could create a material inconsistency;
- the conflict must be resolved by the proper authority;
- all affected downstream artifacts must be identified;
- the resolution must be added to change control; and
- the prior conflicting treatment must remain historically traceable.

Implementation convenience is not authority.

6. PIA Operating Model

Each PIA operates through three controlled baselines.

6.1 As-Designed Baseline

The approved description of what must be built.

It includes:

- requirements;
- workflows;
- data rules;
- permissions;
- technical contracts;
- acceptance criteria;
- evidence requirements; and
- release conditions.

6.2 As-Built Baseline

The actual implementation produced from the approved design.

It includes:

- repository commit;
- source files;
- database migrations;
- infrastructure configuration;
- feature flags;
- environment configuration;
- deployed services;
- generated interfaces;
- integration configuration; and
- known implementation limitations.

6.3 As-Verified Baseline

The implementation state actually tested and supported by evidence.

It includes:

- tested commit or build;
- test environment;
- test results;
- evidence manifest;
- findings;
- deviations;
- unresolved risks;
- deployment evidence; and
- readiness disposition.

The three baselines must remain linked but distinct.

A PIA must not claim that an as-designed requirement is implemented merely because it is described.

A capability must not claim verification when the tested build differs from the identified as-built baseline.

7. Applicability

This Standard applies to:

- foundational platform PIAs;
- business-domain PIAs;
- cross-domain PIAs;
- user-experience PIAs;
- infrastructure PIAs;
- mobile and offline PIAs;
- integration PIAs;
- migration PIAs;
- administrative PIAs;
- AI and automation PIAs;
- onboarding and enrollment PIAs;
- assurance and evidence PIAs; and
- material revisions to an existing approved capability.

Representative PIA families include:

- Identity, Account, Actor, Enrollment, and Onboarding
- Facility, Organization, and Tenancy
- Relationships, Authorization, and Permissions
- Horse Identity, Profile, and Lifecycle

- Health, Welfare, Medical Records, and Clinical Support
- Daily Care Operations
- Feed, Nutrition, Medication, and Treatment Administration
- Scheduling, Calendar, Tasks, and Notifications
- Lessons, Training, Riders, and Guardians
- Shows, Events, Ride Times, and Group Itineraries
- Billing, Payments, Invoicing, and Financial Operations
- Owner Portal and Participant Experience
- Communications, Notices, and Messaging
- Files, Media, Documents, and Digital Assets
- Search and Discovery
- Reporting, Analytics, and Business Intelligence
- AI, Recommendations, and Automation
- Mobile, Offline, and Synchronization
- External Integrations and Developer Platform
- Platform Administration, Support, and Incident Operations

8. PIA Classification

Each PIA must be classified as one or more of the following:

Classification	Purpose
Foundational PIA	Defines capabilities inherited by multiple downstream PIAs.
Domain PIA	Defines one primary business or product domain.
Cross-Domain PIA	Defines a workflow or service spanning multiple authoritative domains.
Platform PIA	Defines shared technical infrastructure or platform operations.
Integration PIA	Defines external adapters, APIs, imports, exports, or webhooks.
Experience PIA	Defines a coherent user experience spanning multiple capabilities.
Migration PIA	Defines controlled conversion, reconciliation, and rollback of data.
Assurance PIA	Defines test, evidence, review, or release-control capabilities.

Combining multiple domains in one PIA is permitted only when:

- the combined capability cannot be coherently implemented separately;
 - authority boundaries remain explicit;
 - the document remains independently reviewable;
 - cross-domain conflicts are not concealed; and
 - the combined scope does not become operationally unmanageable.
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9. BRAVO Quality Standard

Every PIA must satisfy the EquineSync BRAVO standard.

9.1 Buildable

Engineering can implement the capability without inventing material:

- product behavior;
- data ownership;
- permission rules;
- workflow states;
- exception handling;
- release scope; or
- operational controls.

9.2 Reviewable

An independent reviewer can determine whether the PIA is:

- complete;
- internally consistent;
- source-aligned;
- technically coherent;
- safe;
- testable; and
- appropriately scoped.

9.3 Auditable

Material requirements, decisions, changes, exceptions, implementations, tests, findings, and release decisions are traceable and preserved.

9.4 Verifiable

Quality assurance can derive objective tests and determine whether the identified implementation satisfies the PIA.

9.5 Operable

EquineSync can deploy, monitor, support, correct, recover, maintain, and retire the capability.

Failure of any BRAVO element prevents an implementation-readiness disposition.

10. Core Drafting Principles

10.1 No Silent Invention

Material product, security, privacy, safeguarding, permission, data, legal, operational, or release decisions must not be invented by:

- engineers;
- drafting agents;
- reviewers;
- machine validators;
- vendors; or
- automated tools.

Unresolved material matters must be recorded as open decisions.

10.2 No Silent Scope Expansion

A desirable feature is not automatically an approved feature.

New capabilities discovered during drafting must be classified as:

- required by authority;
- necessary implementation detail;
- recommended enhancement;
- deferred capability;
- unresolved proposal; or
- out of scope.

10.3 One Authoritative Owner

Every material record, state, event, and decision must have one authoritative owner.

Other domains may reference, display, cache, transform, or derive information only under expressly defined rules.

10.4 Whole-Workflow Specification

Every important workflow must address:

- initiation;
- validation;
- authorization;
- intermediate states;
- completion;
- cancellation;
- correction;

- reversal;
- expiration;
- failure;
- retry;
- notification;
- audit;
- support; and
- recovery.

10.5 Least Privilege

Access must be limited according to:

- actor;
- role;
- relationship;
- facility;
- organization;
- horse;
- record;
- purpose;
- delegation;
- consent;
- guardian authority; and
- time.

10.6 Evidence by Design

Evidence requirements must be identified before implementation.

10.7 Operational Realism

PIAs must account for use in:

- barns;
- arenas;
- trailers;
- paddocks;
- show grounds;
- veterinary contexts;
- mobile devices;
- poor connectivity;
- outdoor glare;
- urgent situations;
- shared responsibilities;
- temporary delegation;
- multi-horse operations; and
- multi-facility operations.

10.8 Explicit Release Boundaries

Every requirement must identify the milestone for which it is required.

10.9 Human-Readable and Machine-Verifiable

Narrative explanation and structured data must remain materially consistent.

10.10 No False Precision

The PIA must not invent unsupported technical thresholds, legal conclusions, or operational assumptions merely to fill a template.

Where a threshold has not yet been approved, the PIA must define:

- the decision owner;
- the decision method;
- the evidence needed; and
- the gate by which it must be resolved.

11. Controlled Inheritance and Tailoring

PIAs should not duplicate foundational rules unnecessarily.

11.1 Inherited Requirements

A PIA may inherit a requirement from an approved foundational PIA when it records:

- source PIA;
- source requirement identifier;
- inherited version;
- applicability;
- domain-specific extension, if any;
- verification method; and
- effect of later changes.

Inherited requirements remain mandatory.

11.2 Local Extension

A domain PIA may make an inherited rule more restrictive when authorized.

It must not weaken the inherited rule without an approved deviation or higher-authority decision.

11.3 Tailoring Determination

Every required section must be completed or marked:

NOT_APPLICABLE_WITH_JUSTIFICATION

The justification must identify:

- why the section is not applicable;
- which reviewer assessed the determination;
- whether any related requirement is inherited elsewhere; and
- whether the determination creates residual risk.

“No content” is not a valid tailoring decision.

11.4 Shared Control Register

Each PIA must identify shared controls such as:

- authentication;
- authorization;
- audit;
- encryption;
- notifications;
- search permission filtering;
- feature flags;
- deployment;
- backup;
- support access; and
- incident response.

The register must state whether the PIA:

- inherits the control;
- extends the control;
- configures the control;
- tests the control; or
- owns the control.

12. Required Roles and Founder-Led Segregation

Each PIA must identify responsibility for:

Role	Minimum Responsibility
Founder / Approval Authority	Approves scope, material decisions, risks, deviations, implementation authorization, and enrollment readiness.
PIA Owner	Maintains completeness and cross-domain coordination.
Drafting Function	Produces the PIA without unauthorized invention.
Domain Reviewer	Validates operational and domain accuracy.
Architecture Reviewer	Validates technical coherence.
Security and Privacy Reviewer	Validates access and data protection.
Segregated Reviewer	Reviews independently from the drafting pass.
Adversarial Reviewer	Challenges assumptions, misuse cases, and failure behavior.
Machine Validation Function	Validates structure, identifiers, references, and schemas.
Golden-Path Reproduction Function	Reviews realistic end-to-end scenarios.
Evidence Custodian	Preserves and indexes evidence.
Engineering Owner	Converts the PIA into controlled implementation work.
QA Owner	Executes objective verification.
Operational Owner	Confirms support, monitoring, and recovery readiness.

Because EquineSync currently operates with limited internal personnel, one person may perform multiple roles.

When roles overlap, procedural segregation is required through:

- separate drafting and review passes;
- separate controlled prompts or instructions;
- independent source re-reading;
- explicit reviewer-role declaration;
- fresh-context adversarial review;
- evidence review after implementation;
- documented limitations; and
- Founder acknowledgment of the overlap.

No review may be represented as independent when the same work product was merely reread within the drafting pass.

13. PIA Lifecycle

13.1 Lifecycle States

State	Meaning
NOT_STARTED	No controlled work has begun.
SOURCE_ASSEMBLY	Sources are being located and verified.
SCOPE_DEFINITION	Domain boundaries and release scope are being established.
DRAFTING	The PIA is under active drafting.
DESIGN_COMPLETE	All mandatory design sections are populated.
DOMAIN_REVIEW	Domain and operational review is underway.
ARCHITECTURE_REVIEW	Technical and boundary review is underway.
SEGREGATED_REVIEW	Independent review is underway.
ADVERSARIAL_CHALLENGE	Misuse, failure, and ambiguity are being challenged.
MACHINE_VALIDATION	Structural and reference validation is underway.
FOUNDER_REVIEW	Presented for Founder decision.
FOUNDER_APPROVED_DESIGN	The design is approved, but engineering is not necessarily authorized.
IMPLEMENTATION_AUTHORIZED	Engineering may build against the identified baseline.
IN_IMPLEMENTATION	Controlled implementation is underway.
IMPLEMENTED_PENDING_CONFORMANCE_REVIEW	Construction is complete, but design conformity has not been established.
AS_BUILT_RECONCILIATION	The PIA and actual implementation are being reconciled.
VERIFICATION_IN_PROGRESS	Tests and evidence collection are underway.
VERIFIED_WITH_FINDINGS	Verification completed with unresolved findings.
VERIFIED_RELEASE_CANDIDATE	Applicable verification passed.

State	Meaning
OPERATIONALLY_READY	Support, monitoring, deployment, and recovery controls are ready.
READY_FOR_CONTROLLED_FIRST_USER_ENROLLMENT	All applicable enrollment gates are satisfied.
ACTIVE	The PIA controls an operational production capability.
REVISION_REQUIRED	The design requires material revision.
REMEDIATION_REQUIRED	The implementation does not conform and requires correction.
BLOCKED	Progress is blocked.
SUPERSEDED	A later approved version controls.
WITHDRAWN	The PIA is no longer authorized.
ARCHIVED	Retained for history and evidence.

13.2 Design Defect Versus Implementation Defect

A **design defect** exists when the approved PIA is:

- ambiguous;
- incomplete;
- conflicting;
- operationally unsafe;
- technically impossible;
- inconsistent with governance; or
- missing a material decision.

A design defect requires PIA revision.

An **implementation defect** exists when the PIA is adequate but the software does not conform.

An implementation defect requires code, configuration, migration, infrastructure, or test remediation.

The team must not rewrite the PIA merely to make a defective implementation appear conformant.

13.3 Prohibited Status Inflation

A PIA must not be labeled:

- approved when only drafted;
- implementation-ready when material decisions remain open;
- implemented when substantial work remains;

- verified when only reviewed;
- operationally ready when monitoring or recovery is absent;
- enrollment-ready when onboarding, support, or evidence remains incomplete.

The declared status must reflect the least-complete material component.

14. Versioning, Baselining, and Change Control

14.1 Versioning

- **Major:** Material change to scope, authority, data ownership, permissions, architecture, lifecycle, or release posture
- **Minor:** New capability or substantive nonbreaking expansion
- **Patch:** Clarification, reference repair, editorial correction, or non-substantive refinement

14.2 Baseline Freeze

When implementation is authorized, the approved PIA version must be frozen as the implementation baseline.

The implementation authorization record must identify:

- PIA version;
- commit or checksum;
- source baseline;
- unresolved approved P2 risks;
- authorized release scope; and
- implementation start date.

14.3 Change Impact Assessment

Every material proposed change must assess effects on:

- governance;
- MAIP;
- other PIAs;
- requirements;
- workflows;
- data model;
- APIs;
- permissions;
- migrations;
- interfaces;
- tests;
- evidence;
- documentation;

- onboarding;
- support;
- deployment;
- rollback; and
- release timing.

14.4 Emergency Change

Emergency changes may be made only to address an active:

- security incident;
- data-integrity failure;
- safety risk;
- severe outage; or
- regulatory or legal necessity.

Emergency changes must be:

- narrowly scoped;
 - logged;
 - reviewed after implementation;
 - reconciled into the PIA;
 - tested; and
 - approved or reversed promptly.
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15. Implementation Drift Control

Implementation drift occurs when the as-built platform differs from the approved PIA without a controlled change.

Each PIA must include a drift review before verification.

The drift review must compare:

- requirements;
- workflows;
- state models;
- permissions;
- data fields;
- APIs;
- events;
- feature flags;
- user interfaces;
- errors;
- notifications;
- logs;

- administrative controls; and
- release exclusions.

Each difference must be classified as:

- conforming implementation detail;
- approved change;
- PIA clarification needed;
- implementation defect;
- undocumented scope expansion;
- undocumented omission; or
- prohibited deviation.

No release-readiness disposition may be issued while material drift remains unclassified.

16. Identifier Standard

16.1 PIA Identifier

ES-PIA-[DOMAIN]-V[MAJOR].[MINOR].[PATCH]

16.2 Element Identifiers

Element	Format
Requirement	[DOMAIN]-REQ-###
Founder Decision	[DOMAIN]-FD-###
Assumption	[DOMAIN]-ASM-###
Open Question	[DOMAIN]-OQ-###
Workflow	[DOMAIN]-WF-###
Business Rule	[DOMAIN]-BR-###
Entity	[DOMAIN]-ENT-###
State Model	[DOMAIN]-SM-###
Permission Rule	[DOMAIN]-PERM-###
UI Requirement	[DOMAIN]-UI-###
API Contract	[DOMAIN]-API-###
Event	[DOMAIN]-EVT-###
Background Job	[DOMAIN]-JOB-###

Element	Format
Integration	[DOMAIN] - INT - ###
Acceptance Criterion	[DOMAIN] - AC - ###
Test Case	[DOMAIN] - TEST - ###
Evidence Item	[DOMAIN] - EVID - ###
Dependency	[DOMAIN] - DEP - ###
Risk	[DOMAIN] - RISK - ###
Finding	[DOMAIN] - FIND - [P0/P1/P2/P3] - ###
Deviation	[DOMAIN] - DEV - ###
Migration Rule	[DOMAIN] - MIG - ###
Metric	[DOMAIN] - METRIC - ###
Work Package	[DOMAIN] - WP - ###
Rollout Control	[DOMAIN] - ROLLOUT - ###
Support Procedure	[DOMAIN] - SUP - ###

Identifiers must not be reused after retirement.

17. Mandatory PIA Sections

Every PIA must include:

1. Document Control and Status
2. Executive Summary
3. Purpose, Outcomes, and Success Measures
4. Authoritative Sources and Inheritance
5. Scope, Boundaries, and Ownership
6. Definitions and Controlled Vocabulary
7. Actors, Roles, Relationships, and Authorities
8. Capability Map and Release Classification
9. User and Operational Workflows
10. Business Rules and Decision Logic
11. Data Entities, Relationships, and Provenance
12. Record Ownership, Stewardship, Correction, and Retention
13. State and Transition Models
14. Authorization and Permission Matrix
15. User Interface and Experience Requirements
16. API, Event, Job, and Integration Contracts

17. Notifications and Communications
18. Files, Media, and Document Handling
19. Search, Reporting, and Analytics
20. Offline, Device, and Synchronization Behavior
21. Security, Privacy, Consent, Safeguarding, and Abuse Controls
22. AI and Automation Controls
23. Failure Modes, Recovery, Correction, and Reconciliation
24. Observability, Administration, Support, and Incident Operations
25. Nonfunctional and Quality Attribute Requirements
26. Environment, Configuration, Feature Flags, and Secrets Boundaries
27. Migration, Seed Data, and Data Reconciliation
28. Engineering Work Packages and Implementation Sequence
29. Acceptance Criteria
30. Test and Validation Matrix
31. Golden-Path Reproduction Scenarios
32. Adversarial, Negative, and Abuse Scenarios
33. Evidence Requirements, Coverage, and Manifest
34. Deployment, Rollout, Rollback, and Release Controls
35. Enrollment and Onboarding Readiness
36. Dependencies and Critical Path
37. Open Decisions, Assumptions, Findings, Deviations, and Risks
38. Implementation Drift and As-Built Reconciliation
39. Change-Control History
40. Requirement Traceability Matrix
41. Five Mandatory Readiness Questions
42. Review, Approval, Authorization, and Disposition
43. Maintenance, Supersession, and Decommissioning

No required section may be omitted.

18. Requirement Drafting Standard

Each requirement must include:

Field	Required Content
Requirement ID	Stable identifier
Requirement Statement	One normative obligation
Source	Controlling authority or approved implementation basis
Rationale	Why the requirement exists
Actor or System	Performer
Preconditions	Required conditions

Field	Required Content
Required Behavior	Expected outcome
Prohibited Behavior	What must not occur
Failure Behavior	Expected handling
Data Impact	Records read, created, modified, or deleted
Permission Impact	Required authority
Release Classification	Applicable milestone
Acceptance Criteria	Objective proof
Tests	Linked test cases
Evidence	Linked evidence items
Status	Draft, approved, implemented, verified, deferred, or superseded

18.1 Strong Requirement Example

SHOW-REQ-021 : The system SHALL prevent publication of a show schedule containing overlapping ride assignments for the same horse or rider unless a user with show-override authority records a reason and confirms the override.

18.2 Weak Requirement Example

The system should make scheduling easy.

The weak form is prohibited because it is not objectively testable.

19. Source Traceability and Freshness

Every PIA must include a source register.

Each source must identify:

- title;
- source type;
- version;
- date;
- repository path;
- checksum where available;
- authority level;
- supersession status;
- applicable requirements; and

- whether the exact source bytes were reviewed.

Sources must be classified as:

- controlling constitutional authority;
- Founder decision;
- MAIP requirement;
- approved PIA inheritance;
- authoritative technical standard;
- operational evidence;
- assumption;
- recommendation; or
- non-controlling reference.

A source must not be treated as current merely because its file modification date is recent.

The PIA must evaluate source content, authority, and supersession status.

20. Domain Boundaries and Ownership

Each PIA must identify:

- records owned;
- decisions owned;
- records referenced;
- records displayed;
- records derived;
- records cached;
- records prohibited from modification;
- incoming dependencies;
- outgoing dependencies;
- authority conflicts; and
- conflict-resolution authority.

Each material record must identify:

- authoritative system;
- authoritative domain;
- steward;
- creator;
- editor;
- corrector;
- retention authority;
- deletion authority;
- export authority;
- audit requirement; and

- historical preservation requirement.
-

21. Workflow Standard

Every material workflow must define:

1. Identifier
2. Purpose
3. Actors
4. Trigger
5. Preconditions
6. Required inputs
7. Optional inputs
8. Authorization checks
9. Ordered steps
10. State changes
11. Records affected
12. Notifications
13. Audit events
14. Success outcome
15. Partial-success behavior
16. Failure behavior
17. Retry behavior
18. Cancellation
19. Correction
20. Reversal
21. Expiration
22. Offline behavior
23. Accessibility behavior
24. Support path
25. Acceptance criteria
26. Tests
27. Evidence

Flow diagrams may supplement, but must not replace, normative workflow text.

22. State and Transition Standard

Each state model must define:

- permitted states;
- initial state;
- terminal states;
- temporary states;

- suspended states;
- transition triggers;
- authorized actors;
- required data;
- validations;
- audit events;
- notifications;
- reversible transitions;
- prohibited transitions;
- timeout behavior;
- expiration;
- correction; and
- historical preservation.

Material transitions must be enforced in authoritative service logic, not solely in the interface.

23. Authorization and Permission Standard

Permission decisions must account for:

- identity;
- account status;
- role;
- relationship;
- organization;
- facility;
- horse;
- record;
- action;
- purpose;
- delegation;
- guardian authority;
- consent;
- sensitivity;
- time;
- emergency status; and
- platform-administration boundaries.

Permission enforcement must be consistent across:

- web;
- iOS;
- Android;
- API;
- search;
- reporting;

- export;
- notifications;
- background jobs;
- integrations;
- AI;
- administrative tools; and
- support access.

A hidden interface control is not an authorization mechanism.

24. Data and Provenance Standard

Each entity must define:

- purpose;
- owner;
- fields;
- types;
- required status;
- validations;
- relationships;
- uniqueness;
- indexing;
- sensitivity;
- retention;
- correction;
- archival;
- deletion;
- searchability;
- exportability;
- offline availability;
- provenance;
- derivation;
- audit; and
- migration treatment.

The PIA must distinguish:

- authoritative values;
- user-entered observations;
- imported values;
- calculated values;
- inferred values;
- AI-generated suggestions;
- cached values;
- historical snapshots;

- immutable evidence; and
- superseded values.

Derived data must identify the source records and calculation or inference method.

25. User Interface and Experience Standard

Every material interface must define:

- platform;
- actors;
- tasks;
- entry point;
- required fields;
- optional fields;
- validation;
- empty state;
- loading state;
- populated state;
- permission-denied state;
- offline state;
- sync-pending state;
- error state;
- success state;
- destructive-action safeguard;
- accessibility;
- responsive behavior;
- autosave behavior;
- progress preservation;
- user-visible status;
- support path; and
- audit-relevant feedback.

Important actions must clearly show:

- what happened;
 - whether it succeeded;
 - what remains pending;
 - whether synchronization is required;
 - whether another person must act;
 - whether an override occurred; and
 - how an error may be corrected.
-

26. API, Event, Job, and Integration Standard

Each contract must define:

- identifier;
- purpose;
- caller;
- authorization;
- request;
- response;
- validation;
- error handling;
- idempotency;
- retries;
- timeout;
- versioning;
- rate limits;
- privacy;
- logging;
- metrics;
- testing; and
- deprecation behavior.

External integrations must not silently become authoritative sources.

Where an integration is unavailable, fallback and reconciliation behavior must be defined.

27. Offline and Synchronization Standard

Each applicable workflow must define:

- offline read capability;
- offline write capability;
- local retention period;
- device protection;
- queued actions;
- actions prohibited offline;
- stale-data warnings;
- sync trigger;
- retry;
- duplicate prevention;
- conflict detection;
- conflict resolution;
- revocation behavior;
- prolonged-offline behavior;
- user-visible status;

- support diagnostics; and
- data-loss safeguards.

“Offline supported” is not a sufficient requirement.

28. Security, Privacy, Consent, and Safeguarding

Each PIA must address applicable:

- authentication;
- session security;
- least privilege;
- authorization;
- encryption;
- data classification;
- consent;
- legal authority;
- guardian authority;
- minor safeguarding;
- communication protections;
- location protections;
- audit;
- suspicious activity;
- abuse prevention;
- data minimization;
- purpose limitation;
- retention;
- deletion restrictions;
- export restrictions;
- administrative access;
- support access;
- vendor access;
- incident response;
- breach evidence; and
- AI data use.

Sensitive workflows must include misuse and unauthorized-disclosure scenarios.

29. AI and Automation Controls

Any AI or automated function must define:

- intended use;
- prohibited use;
- initiating actor;

- data inputs;
- provenance;
- model or provider boundary;
- permission scope;
- confidence treatment;
- human review;
- correction;
- explanation;
- audit;
- fallback;
- failure;
- monitoring;
- cost controls;
- retention;
- training-data restrictions;
- action authority; and
- kill-switch or disablement method.

AI must not receive broader authority than the initiating user.

AI-generated output must remain visibly distinct from verified authoritative records until appropriately accepted.

30. Failure, Recovery, Correction, and Reconciliation

For every material failure, define:

- cause;
- detection;
- user message;
- system state;
- data impact;
- retry;
- rollback;
- reconciliation;
- escalation;
- alert;
- audit;
- support procedure;
- recovery objective; and
- correction authority.

Required scenarios include:

- invalid input;
- duplicates;

- stale data;
- simultaneous edits;
- unauthorized access;
- failed dependency;
- failed integration;
- failed payment;
- failed upload;
- failed notification;
- partial synchronization;
- job failure;
- timeout;
- configuration error;
- data corruption;
- expired authority;
- transferred horse;
- transferred facility;
- deleted user;
- timezone error; and
- repeated user action.

31. Observability, Support, and Incident Operations

Each PIA must identify:

- service metrics;
- product metrics;
- audit events;
- logs;
- alerts;
- thresholds;
- dashboards;
- ownership;
- escalation;
- health checks;
- failed-job visibility;
- integration monitoring;
- sync monitoring;
- support diagnostic data;
- correction tools;
- replay tools;
- reconciliation tools;
- suspension tools;
- incident severity;
- incident communications; and
- post-incident evidence.

Sensitive data must not be unnecessarily exposed through logs or support tooling.

32. Nonfunctional and Quality Attribute Requirements

Applicable requirements must address:

- availability;
- latency;
- throughput;
- concurrency;
- scalability;
- data integrity;
- resilience;
- recovery;
- accessibility;
- usability;
- compatibility;
- maintainability;
- portability;
- security;
- privacy;
- mobile behavior;
- offline behavior;
- backup;
- retention;
- cost;
- monitoring; and
- deployability.

Thresholds must be measurable or supported by an approved measurement plan.

33. Environment and Configuration Standard

Each PIA must identify behavior and dependencies across:

- local development;
- test;
- integration;
- staging;
- pilot;
- production; and
- disaster-recovery environments, where applicable.

Configuration must identify:

- scope;
- default;
- allowed values;
- environment;
- owner;
- change authority;
- validation;
- audit;
- rollback; and
- secrets boundary.

Secrets must not be included in PIA text, evidence, screenshots, repositories, or logs.

34. Engineering Work Package Standard

An approved PIA must be converted into controlled engineering work packages.

Each work package must include:

- work package ID;
- objective;
- linked PIA requirements;
- affected components;
- dependencies;
- implementation tasks;
- migration tasks;
- test obligations;
- evidence obligations;
- documentation obligations;
- completion criteria;
- reviewer;
- implementation status; and
- repository references.

No work package may be marked complete merely because code was written.

Completion requires satisfaction of its defined implementation, test, evidence, and documentation criteria.

35. Acceptance Criteria

Each acceptance criterion must define:

- identifier;
- requirement;
- precondition;
- actor;
- action;
- expected outcome;
- prohibited outcome;
- audit expectation;
- failure expectation;
- applicable environment;
- release scope; and
- evidence requirement.

Preferred structure:

Given [precondition],
When [action],
Then [required result],
And [required security, audit, data, or notification result],
But not [prohibited result].

36. Test and Validation Standard

Tests must cover applicable:

- units;
- services;
- APIs;
- schemas;
- database constraints;
- states;
- permissions;
- cross-domain integrations;
- interfaces;
- mobile;
- offline;
- synchronization;
- accessibility;
- performance;
- security;
- recovery;
- migration;

- regression;
- golden paths;
- negative paths;
- misuse; and
- abuse cases.

The test matrix must identify the tested:

- PIA version;
- repository commit;
- application build;
- environment;
- configuration profile; and
- data set.

37. Golden-Path Standard

Each PIA must include representative end-to-end scenarios.

At least one golden path must demonstrate the domain's contribution to first-user enrollment.

Golden paths must include:

- realistic actors;
- realistic records;
- starting state;
- ordered actions;
- expected state changes;
- notifications;
- permission decisions;
- audit events;
- integration behavior;
- completion state;
- cleanup; and
- evidence instructions.

38. Adversarial and Negative Review

Each PIA must challenge how the capability could:

- fail;
- be misunderstood;
- be misused;
- be abused;

- expose information;
- bypass authority;
- create duplicate actions;
- create conflicting records;
- generate silent data loss;
- mislead a user;
- harm a minor;
- cause financial error;
- cause horse-health or welfare risk;
- create evidentiary ambiguity;
- fail offline;
- fail under repeated actions; or
- create inappropriate administrative power.

39. Evidence Sufficiency Standard

Evidence must establish:

1. **Identity:** What requirement and test the evidence supports.
2. **Provenance:** Who or what produced it.
3. **Environment:** Where the test occurred.
4. **Baseline:** Which code, build, configuration, and PIA version were tested.
5. **Result:** What occurred.
6. **Integrity:** How alteration or mismatch is prevented or detected.
7. **Limitations:** What the evidence does not prove.
8. **Custody:** Where it is retained.

39.1 Evidence Coverage

Before a verified disposition may be issued:

- every enrollment-blocking requirement must have test coverage;
- every enrollment-blocking acceptance criterion must have evidence;
- every permission family must have positive and negative tests;
- every material state model must have transition tests;
- every material failure path must have validation;
- every critical dependency must have integration evidence; and
- every retained exception must be visible.

39.2 Evidence Package

The final evidence package must contain:

- evidence manifest;
- tested baseline;
- test results;

- findings;
- golden-path results;
- adversarial results;
- migration results;
- deployment results;
- support and recovery results;
- checksums where appropriate; and
- final disposition.

40. Finding Severity

Severity	Meaning
P0	Immediate safety, security, legal, safeguarding, data-integrity, or catastrophic release defect.
P1	Material defect blocking the applicable approval, implementation, verification, or enrollment gate.
P2	Nonblocking weakness or retained risk requiring tracking and acceptance.
P3	Editorial, cosmetic, or low-impact improvement.

No P0 may remain open at implementation authorization.

No relevant P0 or P1 may remain open at first-user enrollment.

41. Release Classification

Each requirement must be classified as:

- REQUIRED_FOR_INTERNAL_BUILD
- REQUIRED_FOR_FOUNDER_PILOT
- REQUIRED_FOR_FIRST_USER_ENROLLMENT
- REQUIRED_BEFORE_PAID_ENROLLMENT
- REQUIRED_FOR_GENERAL_PRODUCTION
- POST_LAUNCH_PLANNED
- DEFERRED_WITH_APPROVAL
- EXPERIMENTAL
- OUT_OF_SCOPE
- PROHIBITED

Deferred work must identify:

- authority;
- rationale;

- risk;
 - user impact;
 - operational impact;
 - target milestone; and
 - re-evaluation date.
-

42. Deployment, Rollout, and Rollback

Each PIA supporting an operational capability must define:

- deployment method;
- environment promotion;
- migration order;
- pre-deployment validation;
- feature-flag treatment;
- rollout cohort;
- pilot limits;
- telemetry;
- stop conditions;
- rollback trigger;
- rollback method;
- data rollback limitations;
- communication plan;
- post-deployment verification; and
- final release acceptance.

A capability may be technically verified but not approved for rollout when operational controls remain incomplete.

43. First-User Enrollment Readiness

A PIA may support first-user enrollment only when:

- all five mandatory readiness questions are **YES_WITH_EVIDENCE** ;
- required capabilities are implemented;
- as-built reconciliation is complete;
- required tests passed;
- golden paths passed;
- adversarial scenarios were addressed;
- evidence is preserved;
- monitoring is active;
- support ownership is assigned;
- correction and recovery tools are ready;
- onboarding instructions exist;

- user terms and consent flows applicable to the PIA are ready;
 - no relevant P0 or P1 remains;
 - retained P2 risks are disclosed and accepted;
 - rollout and rollback are verified; and
 - the Founder issues an enrollment-readiness disposition.
-

44. Five Mandatory Readiness Questions

Question 1

Can engineering build the capability without making unauthorized product decisions?

Question 2

Can quality assurance determine objectively whether the capability works?

Question 3

Can a reviewer trace the capability to EquineSync's controlling governance and the MAIP?

Question 4

Can EquineSync safely operate, support, monitor, recover, and maintain the capability?

Question 5

Can the Founder determine whether the capability is ready for first-user enrollment?

Permitted answers:

- YES_WITH_EVIDENCE
- NO
- PARTIALLY_SATISFIED
- NOT_APPLICABLE_WITH_JUSTIFICATION

Questions 1 through 3 must be YES_WITH_EVIDENCE before implementation authorization.

All five must be YES_WITH_EVIDENCE before first-user enrollment.

45. Gate Model

45.1 Design-Complete Gate

Required:

- sources assembled;
- scope defined;
- mandatory sections populated;
- requirements identified;
- open decisions visible;
- workflows defined;
- ownership and permissions defined;
- acceptance criteria drafted.

45.2 Founder Design-Approval Gate

Required:

- structured reviews completed;
- material conflicts resolved;
- P0 findings closed;
- blocking P1 findings closed;
- design risks disclosed;
- requested Founder decisions identified.

45.3 Implementation-Authorization Gate

Required:

- Questions 1 through 3 are YES_WITH_EVIDENCE ;
- approved baseline frozen;
- engineering work packages prepared;
- dependencies classified;
- test plan complete;
- evidence plan complete;
- change control active.

45.4 Verification Gate

Required:

- implementation complete for applicable scope;
- as-built reconciliation complete;
- tests executed;
- evidence indexed;
- findings classified;
- drift resolved.

45.5 Operational-Readiness Gate

Required:

- monitoring active;
- alerts tested;
- support workflow ready;
- administrative tools ready;
- backup and restore tested;
- rollback tested;
- incident ownership assigned;
- documentation ready.

45.6 First-User Enrollment Gate

Required:

- all five mandatory questions are `YES_WITH_EVIDENCE` ;
 - no relevant P0 or P1 remains;
 - enrollment golden path passes;
 - pilot cohort and limitations defined;
 - onboarding content ready;
 - user-support path active;
 - Founder approval issued.
-

46. Formal Deviation Process

Each deviation must include:

- identifier;
- affected requirement;
- requested departure;
- rationale;
- alternatives considered;
- user impact;
- security impact;
- privacy impact;
- safeguarding impact;
- data impact;
- operational impact;
- testing impact;
- duration;
- compensating control;
- risk owner;
- approval authority;
- expiration; and

- review date.

A deviation must not silently amend locked governance.

47. Required PIA Package

```
/pia-domain/  
PIA_[DOMAIN]_Vx.y.z.md  
PIA_[DOMAIN]_MACHINE_READABLE.json  
SOURCE_REGISTER.md  
INHERITANCE_AND_SHARED_CONTROL_REGISTER.md  
REQUIREMENT_REGISTER.csv  
FOUNDER_DECISION_REGISTER.md  
WORKFLOW_REGISTER.md  
DATA_DICTIONARY.md  
STATE_TRANSITION_MATRIX.csv  
PERMISSION_MATRIX.csv  
API_EVENT_JOB_CONTRACTS.md  
ACCEPTANCE_CRITERIA.csv  
TEST_MATRIX.csv  
GOLDEN_PATHS.md  
ADVERSARIAL_SCENARIOS.md  
ENGINEERING_WORK_PACKAGE_REGISTER.csv  
DEPENDENCY_REGISTER.csv  
RISK_FINDING_DEVIATION_REGISTER.csv  
AS_BUILT_RECONCILIATION.md  
EVIDENCE_MANIFEST.json  
CHANGE_CONTROL_LOG.md  
REVIEW_DISPOSITION.md  
PACKAGE_MANIFEST.json  
CHECKSUM_MANIFEST.sha256
```

Where DOCX or PDF renderings are created, they must remain derived presentation copies unless separately designated as authoritative.

48. Machine-Readable Companion

The machine-readable file must include:

```
{  
  "pia_id": "",  
  "title": "",  
  "version": "",
```

```

"status": "",
"classification": "",
"owner": "",
"source_baseline": [],
"inherited_controls": [],
"requirements": [],
"workflows": [],
"entities": [],
"state_models": [],
"permission_rules": [],
"interfaces": [],
"api_contracts": [],
"events": [],
"jobs": [],
"acceptance_criteria": [],
"tests": [],
"evidence": [],
"work_packages": [],
"dependencies": [],
"risks": [],
"findings": [],
"deviations": [],
"release_scope": [],
"environments": [],
"rollout_controls": [],
"readiness_questions": [],
"approvals": [],
"change_history": []
}

```

Machine validation must confirm:

- required fields are present;
- identifiers are unique;
- references resolve;
- no material requirement is orphaned;
- no test is orphaned;
- no evidence item is orphaned;
- release classifications are valid;
- finding severities are valid;
- readiness answers use permitted values;
- required sections exist;
- package files match the manifest; and
- checksums match where required.

49. Review Protocol

A complete PIA review must include:

1. Source verification
2. Structural completeness
3. Domain review
4. Cross-domain boundary review
5. Architecture review
6. Security and privacy review
7. Permission review
8. Data-integrity review
9. Safeguarding review where applicable
10. Operational-readiness review
11. Adversarial challenge
12. Machine validation
13. Golden-path review
14. Evidence-plan review
15. Founder decision review

Each review must record:

- function;
- reviewer or agent;
- role overlap;
- date;
- version;
- scope;
- findings;
- limitations;
- disposition; and
- required remediation.

50. Maintenance, Supersession, and Decommissioning

Each active PIA must be reviewed when:

- governance changes;
- a related PIA changes materially;
- implementation materially changes;
- a security or privacy incident reveals a gap;
- users reveal a material workflow failure;
- an integration changes;
- a dependency is deprecated;
- a major platform release occurs; or
- the capability is proposed for retirement.

Decommissioning must define:

- replacement capability;
 - user communication;
 - data migration;
 - retention;
 - export;
 - access termination;
 - integration shutdown;
 - feature-flag removal;
 - code removal;
 - evidence preservation; and
 - final archival disposition.
-

PART II

CONTROLLED PIA TEMPLATE

[PIA TITLE]

PIA ID: [ES-PIA-DOMAIN-Vx.y.z]

Version: [x.y.z]

Status: [LIFECYCLE STATUS]

PIA Classification: [FOUNDATIONAL / DOMAIN / CROSS-DOMAIN / PLATFORM / INTEGRATION / EXPERIENCE / MIGRATION / ASSURANCE]

Founder and Approval Authority: Rian Ray

PIA Owner: [NAME OR FUNCTION]

Engineering Owner: [NAME OR FUNCTION]

Operational Owner: [NAME OR FUNCTION]

Evidence Custodian: [NAME OR FUNCTION]

Constitutional Baseline: [COMMIT / TAG]

MAIP Baseline: [VERSION / PATH / HASH]

Repository Path: [PATH]

Effective Date: [DATE OR PENDING]

Release Applicability: [INTERNAL / PILOT / FIRST USER / PAID / GENERAL PRODUCTION]

External Assurance: NOT_EXTERNALLY_ASSURED

1. Document Control and Status

1.1 Control Record

Field	Value
PIA ID	
Title	
Version	
Status	
Classification	
PIA Owner	
Drafting Function	
Review Functions	
Approval Authority	
Source Baseline	
Repository Path	
Effective Date	
Supersedes	
Package Checksum	

1.2 Current Disposition

[DISPOSITION]

1.3 Baseline Status

Baseline	Identifier	Status	Date
As-designed			
As-built			
As-verified			

1.4 Role Segregation Disclosure

Function	Person or Agent	Role Overlap	Procedural Segregation
----------	-----------------	--------------	------------------------

2. Executive Summary

Describe:

- capability;
- users;
- problem;
- release scope;
- critical dependencies;
- major risks;
- current lifecycle state; and
- requested disposition.

2.1 Executive Readiness

Question	Answer	Supporting Sections
Buildable without unauthorized decisions?		
Objectively testable?		
Traceable to governance and MAIP?		
Safely operable and recoverable?		
Ready for first-user enrollment?		

3. Purpose, Outcomes, and Success Measures

3.1 Purpose

[State why the PIA exists.]

3.2 Product Outcome

[Define the user-facing result.]

3.3 Operational Outcome

[Define the operational result.]

3.4 Success Measures

Measure ID	Measure	Target or Decision Method	Data Source	Review Gate
------------	---------	---------------------------	-------------	-------------

3.5 Non-Goals

[List expressly excluded outcomes.]

4. Authoritative Sources and Inheritance

4.1 Source Register

Source ID	Source	Authority	Version / Hash	Exact Source Reviewed?	Requirements Derived
-----------	--------	-----------	----------------	------------------------	----------------------

4.2 Inherited Requirements

Inherited Requirement	Source PIA	Version	Applicability	Local Extension	Verification
-----------------------	------------	---------	---------------	-----------------	--------------

4.3 Shared Control Register

Control	Owning PIA	Treatment	Local Configuration	Local Test
---------	------------	-----------	---------------------	------------

4.4 Founder Decisions

Decision ID	Decision	Date	Status	Requirements Affected
-------------	----------	------	--------	-----------------------

4.5 Source Gaps and Conflicts

ID	Gap or Conflict	Impact	Blocking?	Resolution Authority
----	-----------------	--------	-----------	----------------------

5. Scope, Boundaries, and Ownership

5.1 In Scope

[List included capabilities.]

5.2 Out of Scope

[List excluded capabilities.]

5.3 Domain Ownership

Record, State, or Decision	Ownership Type	Authoritative Domain	Steward	Notes
----------------------------	----------------	----------------------	---------	-------

5.4 Cross-Domain Boundaries

Related Domain	This PIA Owns	This PIA References	Prohibited Overlap	Interface
----------------	---------------	---------------------	--------------------	-----------

5.5 Tailoring Determinations

Required Section	Applicability	Justification	Reviewer
------------------	---------------	---------------	----------

6. Definitions and Controlled Vocabulary

Term	Definition	Source	Prohibited Alternative Meaning
------	------------	--------	--------------------------------

7. Actors, Roles, Relationships, and Authorities

7.1 Actor Register

Actor ID	Actor	Description	Authentication	Typical Authority
----------	-------	-------------	----------------	-------------------

7.2 Relationship Register

Relationship	Parties	Start	End	Authority Created
--------------	---------	-------	-----	-------------------

7.3 Delegation

Delegator	Delegate	Authority	Duration	Revocation	Audit
-----------	----------	-----------	----------	------------	-------

7.4 Guardian and Minor Controls

[Define applicable guardian, consent, communication, location, media, and safeguarding controls.]

8. Capability Map and Release Classification

Capability ID	Capability	Actor	Description	Release Classification	Status
---------------	------------	-------	-------------	------------------------	--------

8.1 Capability Dependencies

[Provide ordered dependencies or a diagram.]

9. User and Operational Workflows

9.X [WORKFLOW ID]: [WORKFLOW NAME]

- Purpose:
- Primary Actor:
- Supporting Actors:
- Trigger:
- Preconditions:
- Required Inputs:
- Optional Inputs:
- Authorization:
- Starting State:
- Completion State:
- Release Classification:

Step	Actor or System	Action	Validation	State Change	Record Impact	Audit	Notification
------	-----------------	--------	------------	--------------	---------------	-------	--------------

- Success Outcome:
- Partial Success:
- Failure Behavior:
- Retry Behavior:
- Cancellation:
- Correction:
- Reversal:
- Expiration:
- Offline Behavior:
- Support Path:
- Acceptance Criteria:
- Tests:
- Evidence:

10. Business Rules and Decision Logic

Rule ID	Rule	Source	Enforcement Layer	Exception Authority	Test
---------	------	--------	-------------------	---------------------	------

11. Data Entities, Relationships, and Provenance

11.1 Entity Register

Entity ID	Entity	Purpose	Owner	Sensitivity	Retention
-----------	--------	---------	-------	-------------	-----------

11.2 Field Register

Entity	Field	Type	Required	Validation	Sensitive	Searchable	Offline
--------	-------	------	----------	------------	-----------	------------	---------

11.3 Relationships

Parent	Child	Relationship	Cardinality	Deletion Behavior
--------	-------	--------------	-------------	-------------------

11.4 Provenance

Record or Field	Classification	Source	Derivation	Human Review	Correction
-----------------	----------------	--------	------------	--------------	------------

12. Record Ownership, Stewardship, Correction, and Retention

Record	Owner	Steward	Creator	Editor	Corrector	Retention	Deletion Authority	Audit
--------	-------	---------	---------	--------	-----------	-----------	--------------------	-------

13. State and Transition Models

13.1 States

State ID	State	Meaning	Entry Requirement	Exit Condition	Terminal?
----------	-------	---------	-------------------	----------------	-----------

13.2 Transitions

From	Trigger	Actor	Required Data	To	Audit	Notification	Reversible
------	---------	-------	---------------	----	-------	--------------	------------

13.3 Prohibited Transitions

[List prohibited transitions and enforcement.]

14. Authorization and Permission Matrix

Permission ID	Actor	Relationship	Resource	Action	Conditions	Delegable	Audit	Notification
---------------	-------	--------------	----------	--------	------------	-----------	-------	--------------

14.1 Platform Administration Boundary

[Define permissible and prohibited platform-administration access.]

14.2 Emergency Access

[Define emergency access or state that it is prohibited.]

14.3 Revocation

[Define revocation timing and enforcement.]

15. User Interface and Experience Requirements

15.1 Supported Surfaces

Surface	Support Level	Uses	Limitations
Web			
iOS			
Android			
Owner Portal			
Admin Portal			
Public Enrollment			

15.2 Interface Register

UI ID	Screen or Component	Actor	Purpose	Required States	Accessibility
-------	---------------------	-------	---------	-----------------	---------------

15.3 Required Interface States

Define:

- initial;
- loading;
- empty;
- populated;
- validation failure;
- permission denied;
- offline;
- sync pending;
- sync failed;
- partial success;
- success;
- destructive confirmation; and
- support escalation.

16. API, Event, Job, and Integration Contracts

16.1 APIs

API ID	Operation	Purpose	Caller	Authorization	Idempotent	Version
--------	-----------	---------	--------	---------------	------------	---------

16.2 Events

Event ID	Event	Producer	Consumers	Payload	Delivery	Audit
----------	-------	----------	-----------	---------	----------	-------

16.3 Jobs

Job ID	Job	Trigger	Schedule	Retry	Escalation	Idempotency
--------	-----	---------	----------	-------	------------	-------------

16.4 Integrations

Integration ID	System	Purpose	Data	Authority	Failure Behavior	Reconciliation
----------------	--------	---------	------	-----------	------------------	----------------

17. Notifications and Communications

Notification ID	Trigger	Recipient	Channel	Content	Timing	Retry	Audit
-----------------	---------	-----------	---------	---------	--------	-------	-------

18. Files, Media, and Document Handling

Asset	Format	Limit	Owner	Access	Retention	Scan	Export
-------	--------	-------	-------	--------	-----------	------	--------

Define upload, versioning, replacement, sharing, deletion, redaction, download, metadata, and evidence preservation.

19. Search, Reporting, and Analytics

19.1 Search

Requirement	Indexed Data	Permission Filter	Ranking	Offline	Audit
-------------	--------------	-------------------	---------	---------	-------

19.2 Reporting

Report	Audience	Data Source	Freshness	Export	Permission
--------	----------	-------------	-----------	--------	------------

19.3 Analytics

[Define metrics, aggregation, privacy, retention, and prohibited inference.]

20. Offline, Device, and Synchronization

Workflow or Record	Offline Read	Offline Write	Queue	Conflict Rule	Status Display	Max Duration
-----------------------	-----------------	------------------	-------	------------------	-------------------	-----------------

Define device encryption, remote revocation, cache expiration, stale data, retries, duplicate prevention, and support diagnostics.

21. Security, Privacy, Consent, Safeguarding, and Abuse Controls

21.1 Data Classification

Data	Classification	Access Basis	Encryption	Retention	Export Restriction
------	----------------	--------------	------------	-----------	--------------------

21.2 Controls

[List applicable security, privacy, consent, and safeguarding controls.]

21.3 Threat and Abuse Register

Threat	Asset	Actor	Control	Detection	Response
--------	-------	-------	---------	-----------	----------

22. AI and Automation Controls

AI ID	Function	Input	Output	Human Review	Authority	Failure	Disablement
-------	----------	-------	--------	--------------	-----------	---------	-------------

Define confidence, provenance, correction, monitoring, privacy, permission boundaries, and prohibited use.

23. Failure Modes, Recovery, Correction, and Reconciliation

Failure ID	Failure	Detection	User Experience	System State	Retry	Recovery	Escalation
------------	---------	-----------	-----------------	--------------	-------	----------	------------

24. Observability, Administration, Support, and Incident Operations

24.1 Metrics

Metric ID	Metric	Purpose	Threshold	Alert	Owner
-----------	--------	---------	-----------	-------	-------

24.2 Logs and Audit

Event	Log Type	Retention	Sensitive Data Treatment	Support Use
-------	----------	-----------	--------------------------	-------------

24.3 Administrative Tools

[List support, correction, suspension, replay, and reconciliation tools.]

24.4 Incident Operations

Severity	Trigger	Owner	Response	Communication
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25. Nonfunctional and Quality Attribute Requirements

NFR ID	Category	Requirement	Threshold or Method	Validation	Release Scope
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26. Environment, Configuration, Feature Flags, and Secrets Boundaries

26.1 Environment Matrix

Environment	Purpose	Data Type	Access	Deployment Method	Restrictions
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26.2 Configuration

Setting	Scope	Default	Values	Authority	Audit	Rollback
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26.3 Feature Flags

Flag	Purpose	Default	Environment	Owner	Removal Condition
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27. Migration, Seed Data, and Reconciliation

27.1 Migration

Migration ID	Source	Target	Mapping	Validation	Rollback	Evidence
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27.2 Seed Data

[List roles, permissions, statuses, categories, templates, and reference values.]

27.3 Reconciliation

[Define record counts, hashes, validation queries, exception handling, and sign-off.]

28. Engineering Work Packages and Implementation Sequence

WP ID	Objective	Requirements	Dependencies	Deliverables	Tests	Evidence	Completion Criteria
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28.1 Sequence

Order	Work Package	Preconditions	Output	Gate
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29. Acceptance Criteria

AC ID	Requirement	Given	When	Then	But Not	Evidence
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30. Test and Validation Matrix

Test ID	Requirement	Test Type	Baseline	Environment	Expected Result	Evidence	Status
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31. Golden-Path Reproduction Scenarios

[GOLDEN PATH ID]: [NAME]

Purpose:

Actors:

Starting State:

Data Set:

Steps:

Expected States:

Expected Records:

Expected Permissions:

Expected Notifications:

Expected Audit Events:

Completion State:

Evidence:

Enrollment Relevance:

32. Adversarial, Negative, and Abuse Scenarios

Scenario ID	Scenario	Risk	Expected Defense	Detection	Evidence
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33. Evidence Requirements, Coverage, and Manifest

33.1 Evidence Manifest

Evidence ID	Requirement or Test	Type	Producer	Baseline	Environment	Integrity	Custodian	Status
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33.2 Coverage Summary

Coverage Area	Required	Covered	Gap	Disposition
Enrollment requirements				
Permissions				
State transitions				
Failure modes				
Dependencies				
Golden paths				

34. Deployment, Rollout, Rollback, and Release Controls

34.1 Deployment

Step	Environment	Action	Validation	Owner	Evidence
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34.2 Rollout

Rollout ID	Cohort	Feature Flag	Telemetry	Stop Condition	Owner
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34.3 Rollback

Trigger	Rollback Action	Data Limitation	Decision Authority	Evidence
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35. Enrollment and Onboarding Readiness

Enrollment Requirement	Owner	Status	Test	Evidence	Blocking
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Include:

- account creation;
- identity verification;
- terms and consent;
- organization or facility enrollment;
- horse onboarding where applicable;
- role assignment;
- permissions;
- welcome communication;
- training;
- support;
- suspension;
- offboarding; and
- recovery.

36. Dependencies and Critical Path

Dependency ID	Dependency	Supplier	Blocking	Version	Verification	Status
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37. Open Decisions, Assumptions, Findings, Deviations, and Risks

37.1 Open Decisions

ID	Question	Impact	Owner	Due Gate	Status
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37.2 Assumptions

ID	Assumption	Basis	Risk	Validation
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37.3 Findings

ID	Severity	Finding	Requirement	Action	Status
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37.4 Deviations

ID	Requirement	Departure	Control	Approval	Expiration
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37.5 Risks

ID	Risk	Likelihood	Impact	Control	Owner	Acceptance
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38. Implementation Drift and As-Built Reconciliation

PIA Element	Approved Design	As Built	Difference	Classification	Resolution
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38.1 As-Built Baseline

- Repository:
- Branch:
- Commit:
- Build:
- Database migration level:
- Infrastructure version:
- Configuration profile:
- Feature flags:
- Deployment environment:

39. Change-Control History

Version	Date	Change	Reason	Impact	Approved By
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40. Requirement Traceability Matrix

Source	Requirement	Workflow	Entity	Permission	Work Package	AC	Test	Evidence	Gate

41. Five Mandatory Readiness Questions

41.1 Engineering Buildability

Answer: [YES_WITH_EVIDENCE / NO / PARTIALLY_SATISFIED]

Basis:

[Explain requirements, workflows, permissions, data, dependencies, and unresolved decisions.]

41.2 Objective QA Verification

Answer: [YES_WITH_EVIDENCE / NO / PARTIALLY_SATISFIED]

Basis:

[Explain acceptance criteria, tests, negative scenarios, golden paths, and evidence.]

41.3 Governance and MAIP Traceability

Answer: [YES_WITH_EVIDENCE / NO / PARTIALLY_SATISFIED]

Basis:

[Explain source and forward-backward traceability.]

41.4 Operational Safety and Recovery

Answer: [YES_WITH_EVIDENCE / NO / PARTIALLY_SATISFIED]

Basis:

[Explain observability, support, recovery, rollback, and maintenance.]

41.5 First-User Enrollment Readiness

Answer: [YES_WITH_EVIDENCE / NO / PARTIALLY_SATISFIED]

Basis:

[Explain enrollment scope, test results, evidence, findings, risks, onboarding, and operational readiness.]

42. Review, Approval, Authorization, and Disposition

42.1 Review Record

Review	Reviewer or Agent	Version	Date	Disposition	Findings
Domain					
Architecture					
Security and privacy					
Segregated review					
Adversarial challenge					
Machine validation					
Golden-path review					
Evidence review					
As-built reconciliation					
Operational readiness					

42.2 Founder Disposition

Select one:

- RETURNED_FOR_REVISION
- FOUNDER_APPROVED_DESIGN_NOT_IMPLEMENTATION_AUTHORIZED
- FOUNDER_APPROVED_AND_IMPLEMENTATION_AUTHORIZED
- FOUNDER_APPROVED_WITH_RETAINED_NONBLOCKING_P2
- IMPLEMENTED_PENDING_CONFORMANCE_REVIEW
- IMPLEMENTATION_REMEDIATION_REQUIRED
- VERIFIED_WITH_OPEN_FINDINGS
- VERIFIED_RELEASE_CANDIDATE
- OPERATIONALLY_READY
- VERIFIED_AND_READY_FOR_CONTROLLED_FIRST_USER_ENROLLMENT
- NOT_READY_FOR_FIRST_USER_ENROLLMENT
- SUPERSEDED
- WITHDRAWN

Founder: Rian Ray

Decision Date:

Disposition:

Conditions:

Retained Risks:

Next Required Gate:

43. Maintenance, Supersession, and Decommissioning

43.1 Review Triggers

[List events requiring reassessment.]

43.2 Supersession

[Identify successor and effective transition.]

43.3 Decommissioning

Area	Required Action	Owner	Evidence
User communication			
Data export or migration			
Retention			
Access termination			
Integration shutdown			
Feature-flag removal			
Code removal			
Evidence archive			

PART III

FOUNDER REVIEW CHECKLIST

A. Authority

- ☐ Controlling sources are verified.
- ☐ Founder decisions are incorporated exactly.
- ☐ Inherited controls are identified.
- ☐ Conflicts are visible.
- ☐ No material requirement was silently invented.

B. Scope

- ☐ In-scope capabilities are explicit.

- ☐ Out-of-scope capabilities are explicit.
- ☐ Release classifications are assigned.
- ☐ Domain ownership is clear.
- ☐ Cross-domain overlap is controlled.
- ☐ Tailoring decisions are justified.

C. Product and Operations

- ☐ Actors are defined.
- ☐ Relationships are defined.
- ☐ Workflows are complete.
- ☐ Negative paths are addressed.
- ☐ Cancellation, correction, and reversal are addressed.
- ☐ Offline behavior is defined.
- ☐ Support paths are defined.

D. Data and Authority

- ☐ Entities are defined.
- ☐ Provenance is defined.
- ☐ Ownership and stewardship are defined.
- ☐ Permission rules are explicit.
- ☐ Delegation and revocation are defined.
- ☐ Guardian and minor controls are addressed.

E. Technical Implementation

- ☐ APIs, events, and jobs are defined.
- ☐ Integrations and failure behavior are defined.
- ☐ Environments are defined.
- ☐ Configuration and feature flags are controlled.
- ☐ Work packages are traceable to requirements.
- ☐ Migration and reconciliation are addressed.

F. Security and Resilience

- ☐ Authentication and authorization are defined.
- ☐ Privacy and consent are addressed.
- ☐ Safeguarding is addressed.
- ☐ Abuse cases are addressed.
- ☐ Monitoring and alerts are defined.
- ☐ Backup, restore, and rollback are defined.
- ☐ Administrative access is bounded.

G. Verification and Evidence

- ☐ Acceptance criteria are objective.

- [] Tests map to requirements.
- [] Positive and negative tests exist.
- [] Golden paths exist.
- [] Adversarial scenarios exist.
- [] Evidence requirements are defined.
- [] Coverage gaps are visible.
- [] Evidence custody is assigned.

H. Release and Enrollment

- [] The as-built baseline is identified.
- [] Drift has been reviewed.
- [] Deployment controls are defined.
- [] Rollout and stop conditions are defined.
- [] Rollback is verified.
- [] Onboarding requirements are complete.
- [] Support is ready.
- [] No relevant P0 or P1 remains.
- [] Retained P2 risks are disclosed.
- [] All five readiness questions are answered.

PART IV

APPROVAL EFFECT

Upon Founder adoption, this Standard becomes the controlling standard for all EquineSync PIAs.

No EquineSync PIA may be represented as:

- design-complete;
- implementation-ready;
- implementation-authorized;
- implemented;
- conformant;
- verified;
- operationally ready;
- release-ready; or
- ready for first-user enrollment

unless it satisfies the applicable requirements and gate criteria established by this Standard or is covered by a valid Founder-authorized deviation.

The controlling implementation principle is:

Governance establishes authority.
The MAIP establishes coordinated implementation direction.
The PIA establishes domain-level executable truth.
Engineering creates the as-built implementation.
Reconciliation detects drift.
Testing challenges the implementation.
Evidence proves the verified result.
Operations make the capability supportable and recoverable.
The Founder determines whether enrollment may begin.